PROJECT TITLE

For

Sensor Lab mini Project

Third Year (Semester-VI) Bachelors in Information Technology, Autonomous Program

By

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Certificate

This is to certify that **Siddhant Sathe (50), Arnav Sawant (52), and Pranav Titambe (60)** have completed the project report on the topic **ESP8266 RFID Attendance System with Google Sheets** satisfactorily in partial fulfilment of the requirements for the award of Mini Project in sensor lab of third Year, (Semester-VI) in Information Technology under the guidance of Mr. Abhishek Chaudhari during the year 2023-2024 as prescribed by An Autonomous Institute Affiliated to University of Mumbai.

Supervisor/ Examiner

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Abstract

The ESP8266 RFID Attendance System is an Internet of Things (IoT) based solution designed to automate attendance tracking in educational institutions and workplaces. The system utilizes a NODEMCU ESP8266 microcontroller, an RC522 RFID reader module, and Google Sheets integration to provide a paperless, efficient method of recording attendance. When an authorized RFID tag is scanned, the system identifies the individual, logs their attendance with timestamp information, and uploads this data in real-time to Google Sheets via Wi-Fi connectivity. The system features a user-friendly interface with an LCD display that provides immediate feedback and a buzzer for audio confirmation. This project demonstrates the practical application of IoT concepts in solving everyday administrative challenges while offering a cost-effective alternative to traditional attendance systems.

Chapter 1

Introduction

The ESP8266 RFID Attendance System is an IoT-based project that utilizes the NODEMCU ESP8266 microcontroller, an RC522 RFID reader, and Google Sheets for real-time data logging. This system is designed to streamline attendance tracking by scanning RFID tags associated with individuals. The scanned data is automatically sent to a Google Sheet via Wi-Fi, eliminating the need for manual record-keeping.

1.1 Overview

Traditional attendance systems often involve manual processes that are time-consuming, prone to errors, and difficult to analyze. The ESP8266 RFID Attendance System addresses these issues by providing an automated solution that offers real-time tracking, accurate data collection, and easy accessibility. This IoT-based approach represents a significant improvement over conventional methods, enhancing efficiency and reliability in attendance management.

1.2 Working Principle

The system operates based on the following principles:

1. **RFID Tag Scanning**: When an RFID tag is brought near the RC522 RFID reader, it detects the unique ID of the tag.
2. **Data Processing**: The NODEMCU ESP8266 processes this ID and associates it with a predefined user in the system.

Chapter 2

Review of Literature

Survey

1) RFID-based Attendance Management System.
   a) This study provides a comprehensive review of RFID technology in attendance systems, highlighting advantages over traditional methods.
2) IoT Applications in Education:
   a) The authors analyze various IoT-based attendance systems, focusing on cloud integration and real-time data processing capabilities.
3) Comparative Analysis of Biometric and RFID-based Authentication Systems
   a) This paper compares RFID technology with biometric solutions, evaluating cost-effectiveness and implementation challenges.

Research Gap

While existing RFID attendance systems offer numerous advantages, many solutions lack:

1. Seamless cloud integration with easily accessible platforms like Google Sheets
2. Low-cost implementation using widely available components
3. Real-time feedback mechanisms for users
4. Open-source architecture that allows for customization and expansion

Our project addresses these gaps by implementing an affordable solution with direct Google Sheets integration, providing immediate feedback through an LCD display and buzzer, and utilizing open-source hardware and software components.

Chapter 3

Project Description

3.1 Problem Definition

Manual attendance systems face several challenges:

- Time-consuming processes that reduce productive classroom or workplace time
- Susceptibility to proxy attendance and human errors
- Difficult data management and report generation
- Lack of real-time tracking and monitoring capabilities

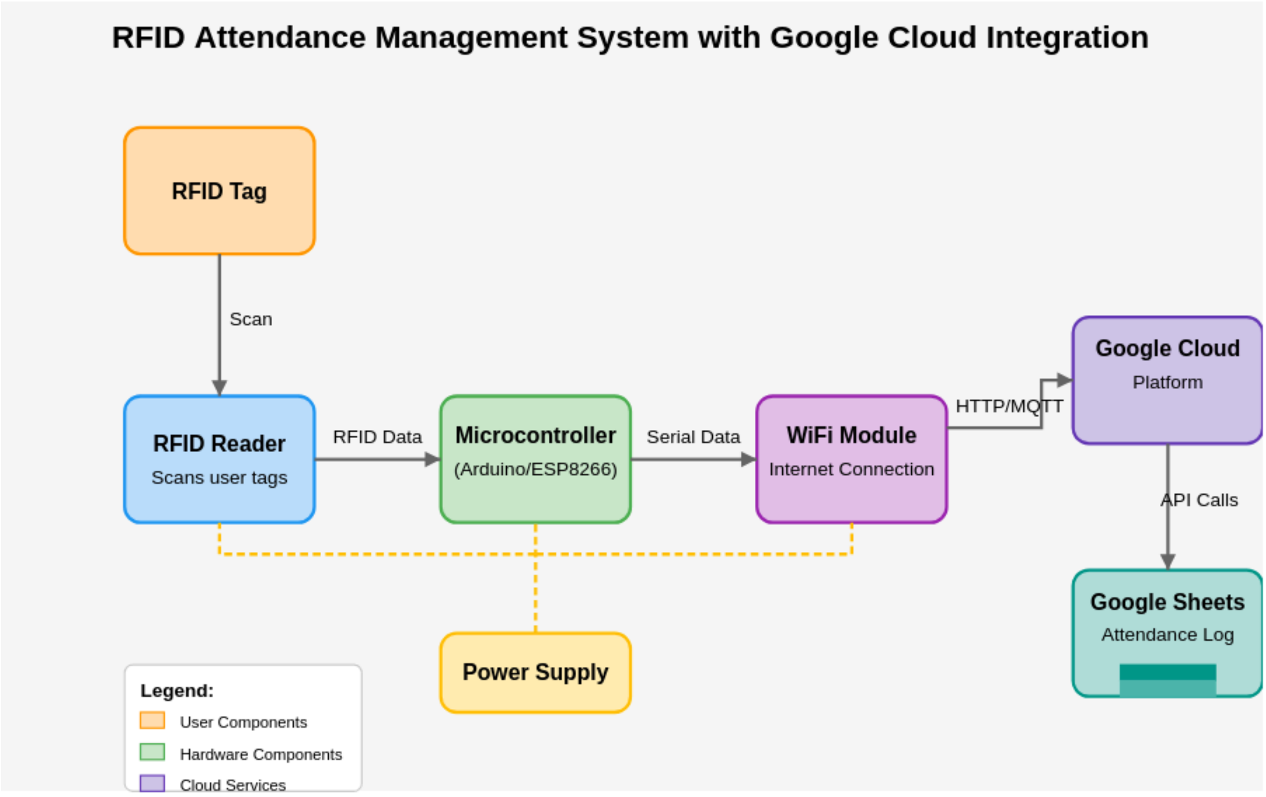
The ESP8266 RFID Attendance System aims to overcome these challenges by providing an automated, reliable, and efficient solution for attendance tracking that leverages IoT technology.

3.2 Steps Involved

The implementation of the ESP8266 RFID Attendance System involves the following steps:

1. **System Design**: Planning the architecture, selecting appropriate components, and defining the system workflow.
2. **Hardware Assembly**: Connecting the NODEMCU ESP8266, RC522 RFID reader, LCD display, and buzzer according to the circuit diagram.
3. **Software Development**: Programming the ESP8266 using Arduino IDE to handle RFID scanning, Wi-Fi connectivity, and data transmission.
4. **Google Sheets Integration**: Setting up Google Apps Script to receive and process data sent from the ESP8266.
5. **Testing and Calibration**: Verifying system functionality, resolving issues, and optimizing performance.
6. **Deployment**: Installing the system in the intended environment and training users on operation.

3.3 Block Diagram



3.4 Component Description/ Hardware & Software

3.4.1 Hardware Components

1. **NODEMCU ESP8266**:
    - Functions as the central controller
    - Provides Wi-Fi connectivity for data transmission
    - Processes RFID tag information
    - Specifications: 80MHz CPU, 4MB flash memory, 3.3V operating voltage
2. **RC522 RFID Reader**:
    - Detects and reads RFID tags
    - Operating frequency: 13.56MHz
    - Reading distance: 0-50mm depending on the tag
    - Interfaces with ESP8266 via SPI protocol
3. **16x2 I2C LCD Display**:
    - Provides visual feedback about system status
    - Displays messages such as "Scan your Card" or "Data Recorded"
    - I2C interface reduces the number of pins required for connection
4. **RFID Tags**:
    - Passive tags containing unique identification information
    - No battery required, powered by electromagnetic field from reader
    - Typically in card or key fob form factor
5. **Buzzer**:
    - Provides audio feedback for successful or unsuccessful scans
    - Operating voltage: 3.3-5V
6. **Breadboard & Jumper Wires**:

- Used for prototyping and connecting components

3.4.2 Software Components

1. **Arduino IDE**:
    - Used for programming the NODEMCU ESP8266
    - Provides libraries for RFID, LCD, and Wi-Fi functionality
2. **Google Apps Script**:
    - Facilitates communication between the ESP8266 and Google Sheets
    - Processes incoming data and updates the spreadsheet
3. **Libraries Used**:
    - SPI.h: For SPI communication with the RFID reader
    - MFRC522.h: For interfacing with the RC522 RFID reader
    - ESP8266WiFi.h: For Wi-Fi connectivity
    - ESP8266HTTPClient.h: For making HTTP requests
    - LiquidCrystal_I2C.h: For controlling the I2C LCD display

3.5 Working of the project

The ESP8266 RFID Attendance System operates through the following sequence:

1. **Initialization**:
    - The system boots up and establishes Wi-Fi connectivity
    - The LCD displays "Initializing" followed by "Scan your Card"
    - The RFID reader is activated and ready to detect tags
2. **Tag Detection**:
    - When an RFID tag is brought near the reader, the system detects it
    - The RC522 reader communicates the tag's unique ID to the ESP8266
3. **User Identification**:
    - The ESP8266 processes the tag ID and retrieves the associated user name
    - The LCD displays "Hey [Name]!" acknowledging the user
4. **Data Transmission**:
    - The ESP8266 establishes a secure HTTPS connection to the Google Script URL
    - It sends the user information along with the current timestamp
5. **Confirmation**:
    - Upon successful data transmission, the LCD displays "Data Recorded"
    - The buzzer provides audio feedback (two short beeps)
    - The system returns to the ready state, displaying "Scan your Card"

3.5.1 Working Flow

1. The system continuously checks for new RFID cards
2. When a card is detected, it reads the stored data

3. The system authenticates the card using a security key
4. If authentication is successful, it reads the user information
5. The ESP8266 connects to the configured Wi-Fi network
6. It sends the user data to Google Sheets via HTTPS
7. The system provides visual and audio feedback to confirm successful recording
8. It returns to the idle state, ready for the next scan

Chapter 4

Implementation

4.1 Hardware

The hardware implementation involves connecting the components according to the following pinout diagrams:

NODEMCU ESP8266 to RC522 RFID Reader

RC522 Pin NODEMCU Pin

VCC	3.3V
GND	GND
SDA	D2
SCK	D5
MOSI	D7
MISO	D6
RST	D4

NODEMCU ESP8266 to LCD I2C

LCD I2C Pin NODEMCU Pin

VCC	3.3V
GND	GND
SDA	D1
SCL	D2

NODEMCU ESP8266 to Buzzer

Buzzer Pin NODEMCU Pin

Positive	D8
Negative	GND

The physical implementation is shown in the following image:

4.2 Software (Flowchart/ Algorithms)

Main Program Flowchart:

1. Start
2. Initialize LCD, SPI, RFID reader, and buzzer
3. Connect to Wi-Fi network
4. Display "Scan your Card" on LCD
5. Check if new card is present
   - If no card detected, go back to step 5
   - If card detected, proceed to next step
6. Read data from RFID card
7. Display user name on LCD
8. Provide audio feedback through buzzer
9. Check Wi-Fi connection status
   - If connected, proceed to next step
   - If not connected, display error and go back to step 4
10. Send data to Google Sheets via HTTPS
11. Check response code
    - If successful, display "Data Recorded" on LCD
    - If failed, display error message
12. Wait for 2 seconds
13. Go back to step 4

RFID Reading Algorithm:

Function ReadDataFromBlock(blockNum, readBlockData[]):

1. Initialize security key (all bytes set to 0xFF)

2. Authenticate the block for read access using Key A

- If authentication fails, display error and return

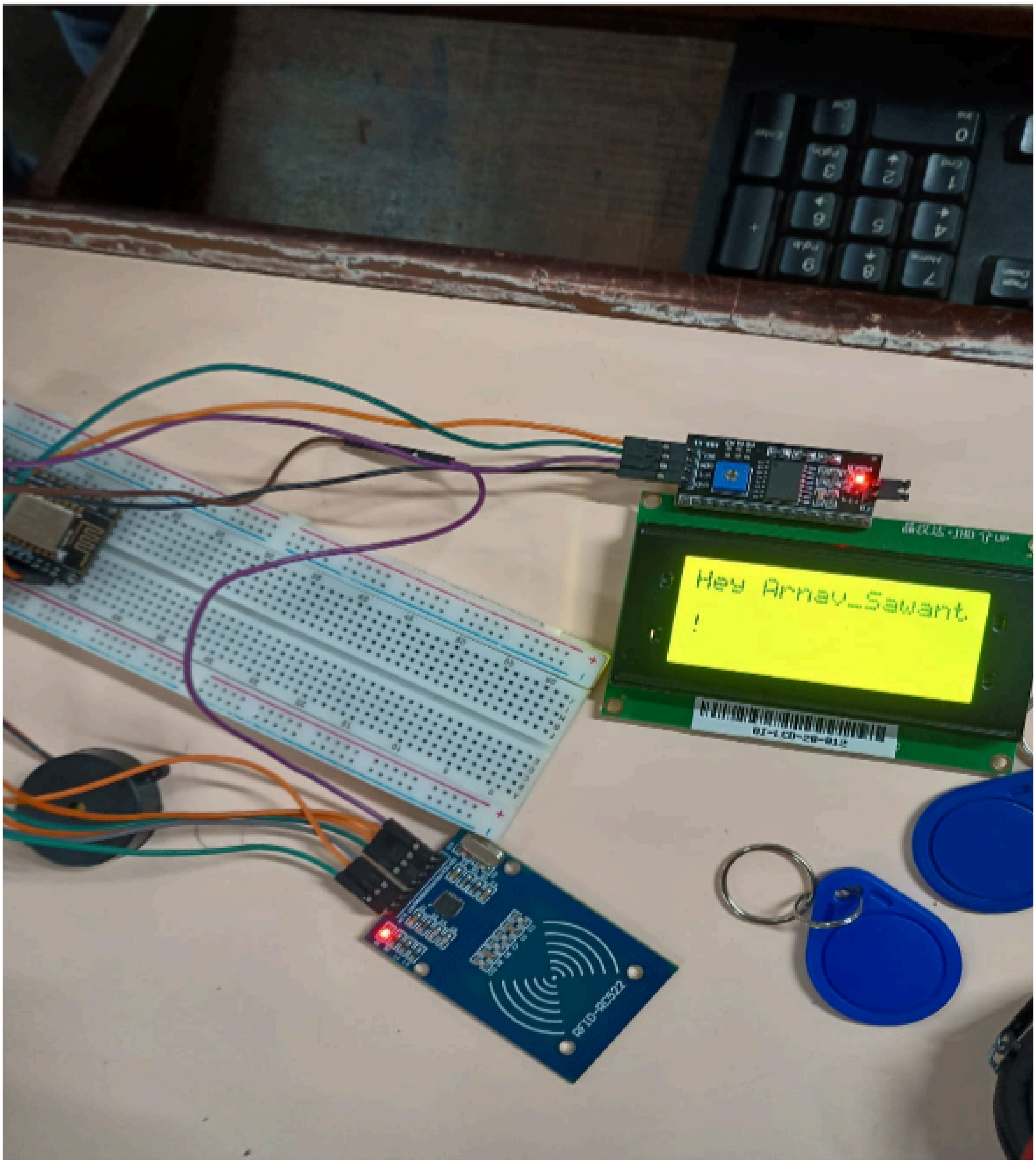
- If authentication succeeds, proceed

3. Read data from the specified block

- If reading fails, display error and return

- If reading succeeds, return block data

End Function



Date	Time	Name
3/26/2025	11:13:50	Arnav_Sawant
3/26/2025	11:14:01	Siddhant_Sathe
3/26/2025	11:16:31	Sung_Jinwoo
3/26/2025	11:16:46	Arnav_Sawant
3/26/2025	11:17:11	Pranav_Pol
3/26/2025	11:17:22	Pranav_Titambe
3/26/2025	11:20:35	Sung_Jinwoo
3/26/2025	11:21:29	Arnav_Sawant
3/26/2025	11:21:42	Pranav_Pol
3/26/2025	11:21:53	Siddhant_Sathe
3/26/2025	11:27:00	Pranav_Titambe
3/26/2025	11:32:48	Ram
4/9/2025	11:08:24	Siddhant_Sathe
4/9/2025	11:09:53	Sung_Jinwoo
4/9/2025	11:10:11	Siddhant_Sathe
4/9/2025	11:10:22	Siddhant_Sathe
4/9/2025	11:10:37	Arnav_Sawant
4/9/2025	11:16:17	Pranav_Pol
4/9/2025	11:22:30	Sung_Jinwoo
4/9/2025	11:31:21	Arnav_Sawant

Chapter 5

Results and Conclusion

Results & Observations

The ESP8266 RFID Attendance System was tested in a simulated environment, and the following observations were recorded:

- **RFID Tag Detection**: The RC522 RFID reader successfully scanned RFID tags, accurately identifying their unique IDs.
- **Real-Time Data Logging**: The NODEMCU ESP8266 transmitted attendance data to Google Sheets without delays, ensuring real-time updates.
- **LCD Display Feedback**: The 16x2 I2C LCD provided clear feedback messages such as "Attendance Marked" or "Unknown Tag," enhancing user interaction.
- **Buzzer Feedback**: The buzzer emitted distinct sounds upon successful scans, providing audible confirmation of attendance marking.
- **Wi-Fi Connectivity**: The ESP8266 maintained stable Wi-Fi communication throughout the testing phase, ensuring seamless data transmission.

Conclusion

The ESP8266 RFID Attendance System is a practical and efficient IoT-based solution for automating attendance tracking. By leveraging the ESP8266 microcontroller, RFID technology, and Google Sheets integration, this project demonstrated how modern technology can simplify and streamline attendance management processes. The system successfully achieved real-time data logging, accurate identification of individuals, and user-friendly interaction through an LCD display and buzzer feedback.

The implementation provides several advantages over traditional attendance systems:

1. Elimination of manual record-keeping and associated errors
2. Real-time data accessibility from anywhere through Google Sheets
3. Reduced time spent on administrative tasks
4. Enhanced accuracy in attendance tracking
5. Cost-effective solution using readily available components

Future enhancements could include adding a web interface for administration, implementing additional authentication mechanisms, and integrating with other educational or workplace management systems.

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